WORKED SOLUTION

STEP III 1996, Question 8

Pure • Möbius Transformations

STEP III 1996, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q8

The key idea

These are Möbius (fractional linear) transformations, and almost everything follows from two observations. Clearing the denominator in $y = T(x)$ gives an equation that is linear in x *and* linear in y, so it can be solved for either one. And writing $T(x)$ as a constant plus a multiple of $1/(cx-d)$ shows at once which value T never takes. For the composition, just substitute and multiply through; the result is easiest to organise by matching each T with the matrix $\begin{pmatrix} a & -b \\ c & -d \end{pmatrix}$.

Solution

Fixed points

For $x \neq d/c$, $T(x) = x$ means $ax - b = x(cx - d)$, that is,

$$cx^2 - (a+d)x + b = 0. \qquad (*)$$

First we check that $(*)$ cannot produce the forbidden value $x = d/c$ (when $c \neq 0$). Substituting gives $d^2/c - (a+d)d/c + b = (bc - ad)/c \neq 0$, so every root of $(*)$ is a genuine fixed point.

If $c \neq 0$, $(*)$ is a quadratic, and the fixed points are

$$x = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4bc}}{2c}.$$

There are two of them if $(a+d)^2 > 4bc$, one if $(a+d)^2 = 4bc$, and none if $(a+d)^2 < 4bc$.

If $c = 0$, the condition $ad \neq bc$ becomes $ad \neq 0$, so $d \neq 0$ and $T(x) = (b - ax)/d$ is defined everywhere. Now $(*)$ reads $(a+d)x = b$. If $a + d \neq 0$ there is exactly one fixed point, $x = b/(a+d)$. If $a + d = 0$ and $b \neq 0$ there are none. If $a + d = 0$ and $b = 0$, then $T(x) = -ax/d = x$, so T is the identity and every x is fixed.

The inverse

From $y(cx-d) = ax-b$ we get $cxy - dy = ax - b$, so $x(cy-a) = dy - b$, and therefore

$$x = T^{-1}(y) = \frac{dy-b}{cy-a}.$$

This has the same form as T, with

$$a' = d, \qquad b' = b, \qquad c' = c, \qquad d' = a,$$

and the condition still holds because $a'd' - b'c' = da - bc \neq 0$. (Dividing by $cy - a$ is legitimate, because $y = a/c$ is never a value of T; see the next part.)

The sets S_r

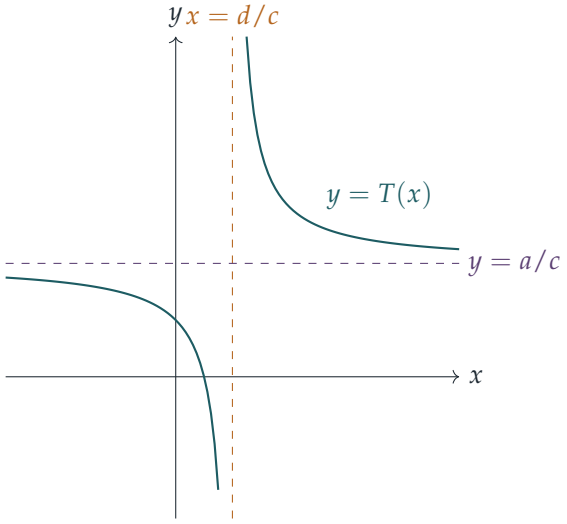
Suppose $c \neq 0$. $T(x)$ is defined exactly when $cx \neq d$, so we take $r = d/c$. Then T is defined for every $x \in S_{d/c}$. To find the image, we split off the constant part:

$$ax - b = \frac{a}{c}(cx-d) + \frac{ad-bc}{c}, \qquad \text{so} \qquad T(x) = \frac{a}{c} + \frac{ad-bc}{c\,(cx-d)}.$$

Since $ad - bc \neq 0$, the second term is never zero, so $T(x) \neq a/c$. Conversely, every $y \neq a/c$ is attained. The inverse formula gives $x = (dy-b)/(cy-a)$, and this x is not d/c, because d/c is not a value of T^{-1} (apply the same argument to T^{-1}, whose "a'/c'" is d/c). Hence

$$T(S_{d/c}) = S_{a/c},$$

and T is a bijection from $S_{d/c}$ to $S_{a/c}$. The graph is a rectangular hyperbola with asymptotes $x = d/c$ and $y = a/c$:



(The sketch uses $T(x) = (2x-1)/(x-1) = 2 + 1/(x-1)$.)

If $c = 0$, then $d \neq 0$ and $a \neq 0$ (because $ad \neq 0$), and $T(x) = (b - ax)/d$ is a non-constant linear function. So T is defined for *all* real x, with no value to exclude, and it maps $\mathbb{R}$ onto $\mathbb{R}$ bijectively. If we still remove a point r, then $T(S_r) = S_{T(r)} = S_{(b-ar)/d}$.

Composition

Substitute $T_1(x) = \dfrac{a_1x - b_1}{c_1x - d_1}$ into T_2, and multiply the top and bottom by $c_1x - d_1$:

$$\begin{aligned} T_2(T_1(x)) &= \frac{a_2(a_1x - b_1) - b_2(c_1x - d_1)}{c_2(a_1x - b_1) - d_2(c_1x - d_1)} \\ &= \frac{(a_1a_2 - b_2c_1)\,x - (a_2b_1 - b_2d_1)}{(a_1c_2 - c_1d_2)\,x - (b_1c_2 - d_1d_2)}. \end{aligned}$$

So T_3 has the same form, with

$$a_3 = a_1a_2 - b_2c_1, \quad b_3 = a_2b_1 - b_2d_1, \quad c_3 = a_1c_2 - c_1d_2, \quad d_3 = b_1c_2 - d_1d_2.$$

? **Ask yourself** Is "the same form" really proved? We also need $a_3d_3 \neq b_3c_3$.

Yes, we do need it. These coefficients are exactly the entries of the matrix product

$$\begin{pmatrix} a_2 & -b_2 \\ c_2 & -d_2 \end{pmatrix} \begin{pmatrix} a_1 & -b_1 \\ c_1 & -d_1 \end{pmatrix} = \begin{pmatrix} a_3 & -b_3 \\ c_3 & -d_3 \end{pmatrix},$$

and each matrix has determinant $bc - ad$. Since the determinant of a product is the product of the determinants,

$$b_3c_3 - a_3d_3 = (b_1c_1 - a_1d_1)(b_2c_2 - a_2d_2) \neq 0.$$

(You can also expand directly; it is a few lines of algebra.) So T_3 is a transformation of the same type.

When is T^2 the identity?

Take $T_1 = T_2 = T$ in the composition formula. Then $T^2(x) = \dfrac{Ax - B}{Cx - D}$ with

$$A = a^2 - bc, \qquad B = b(a - d), \qquad C = c(a - d), \qquad D = bc - d^2.$$

T^2 is the identity when $Ax - B = x(Cx - D)$ for all x in its domain, which is all but at most a few points. A polynomial identity holding at infinitely many points holds identically, so we compare coefficients:

$$C = 0, \qquad A + D = 0, \qquad B = 0,$$

that is,

$$c(a - d) = 0, \qquad (a - d)(a + d) = 0, \qquad b(a - d) = 0.$$

- If $a = d$, all three hold. Indeed $T^2(x) = \dfrac{(a^2 - bc)x}{-(bc - a^2)} = x$, where $a^2 - bc = ad - bc \neq 0$.
- If $a \neq d$, we need $b = c = 0$ and $a = -d$. Then $T(x) = ax/(-d) = x$, so T is the identity itself.

So T^2 is the identity **if and only if** $a = d$**, or** $b = c = 0$ **and** $a + d = 0$ (that is, T is the identity). The second case is trivial, so the interesting involutions are exactly the maps $x \mapsto \dfrac{ax - b}{cx - a}$ with $a^2 \neq bc$.

An example

By the condition just found, we need two maps with $a = d$ whose composition has $a \neq d$. Take

$$T_1(x) = -x \quad (a_1, b_1, c_1, d_1) = (1, 0, 0, 1), \qquad T_2(x) = 1 - x \quad (a_2, b_2, c_2, d_2) = (1, 1, 0, 1).$$

Each has $a = d$, so T_1^2 and T_2^2 are the identity. You can see this directly too: $-(-x) = x$ and $1 - (1 - x) = x$. Their composition is

$$T_3(x) = T_2(T_1(x)) = 1 - (-x) = x + 1,$$

which in the standard form has $(a_3, b_3, c_3, d_3) = (1, -1, 0, -1)$, so $a_3 \neq d_3$. Indeed $T_3^2(x) = x + 2 \neq x$.

Geometrically, two reflections of the line (in 0 and in $\frac{1}{2}$) compose to a translation, and a translation is not its own inverse. For an example with $c \neq 0$, try $T_1(x) = -1/x$ with the same T_2. Then $T_3(x) = (x + 1)/x$, and $T_3^2(x) = (2x + 1)/(x + 1)$.

Checks

- Use the sketched example $T(x) = (2x - 1)/(x - 1)$ ($a = 2$, $b = 1$, $c = d = 1$). Its fixed points are the roots of $x^2 - 3x + 1 = 0$, namely $x = \frac{1}{2}(3 \pm \sqrt{5})$, and you can check that $T(2.618\ldots) = 2.618\ldots$. Its inverse is $(y - 1)/(y - 2)$, which is undefined at $y = 2 = a/c$, as predicted.
- The composition formula with $T_1 = T$, $T_2 = T^{-1}$, i.e. $(a_2, b_2, c_2, d_2) = (d, b, c, a)$, gives $a_3 = ad - bc$, $b_3 = 0$, $c_3 = 0$, $d_3 = bc - ad$. This is the identity, as it should be.
- A computer verified the inverse, image and composition formulae, using exact fractions on thousands of random (a, b, c, d). It also tested the condition for T^2 to be the identity against direct evaluation for every integer choice of $a, b, c, d \in [-3, 3]$ with $ad \neq bc$.

Common slips

- Forgetting the trivial case: $a = d$ is not *necessary*, because the identity ($b = c = 0$, $a = -d$) also squares to the identity.

- Showing the composition is a ratio of linear functions but not checking $a_3d_3 \neq b_3c_3$.
- In the $c = 0$ case, looking for an excluded point. A non-constant linear map is defined everywhere and hits every real number.

Beyond STEP

Allowing $x = \infty$, with $T(d/c) = \infty$ and $T(\infty) = a/c$, makes every such T a bijection of the extended line. With complex coefficients you get the Möbius transformations of the Riemann sphere. The matrix correspondence used above says that these maps form a group, isomorphic to invertible 2×2 matrices modulo scalars (the projective linear group). Composition becomes matrix multiplication, and a map other than the identity is an involution exactly when its matrix has trace zero; here that is $a - d = 0$. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.